

CYBERSECURITY ANALYST PORTFOLIO CASE STUDY

Network Reconnaissance and Service Exposure

Authorized subnet discovery, full TCP inventory, service fingerprinting, firewall hardening, validation, and evidence integrity

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OUTCOME Discovered four active lab hosts, identified only SSH and HTTP on the authorized Ubuntu target, reduced firewall source scope from Anywhere to the isolated lab subnet, confirmed TCP/8080 remained filtered, and verified eight evidence files with SHA-256.

Component	Address / role	Primary use
Parrot Security 7.3	10.10.10.5 - analyst workstation	Nmap 7.95 and WhatWeb
Ubuntu Server 24.04	10.10.10.4 - authorized target	SSH and Apache
SOC-Lab	10.10.10.0/24 isolated NAT network	Controlled assessment boundary

Authorized local simulation only. No external or production system was targeted. AI assistance is disclosed in this report.

1. Executive Summary

I performed an authorized network reconnaissance and service-exposure assessment from Parrot Security against LINUX-WEBSERVER01 at 10.10.10.4. Subnet discovery identified four active lab addresses. A complete TCP SYN scan of all 65,535 ports found only SSH on TCP/22 and HTTP on TCP/80; the remaining 65,533 ports were filtered.

Targeted fingerprinting identified OpenSSH 9.6p1 Ubuntu and an Apache HTTP service. WhatWeb confirmed Apache and the previously deployed response-security headers while detailed Apache version disclosure remained suppressed. I correlated the external scan with target-side listener and service checks, then reviewed UFW policy.

The initial firewall allowed SSH and HTTP from Anywhere. I added subnet-scoped rules for 10.10.10.0/24 and removed the broad rules. Post-change scanning confirmed authorized access to TCP/22 and TCP/80 remained available while TCP/8080 remained filtered. Eight evidence artifacts were recorded and verified against a SHA-256 manifest.

Measure	Result	Analyst meaning
Active lab hosts	4	Known virtual infrastructure and systems enumerated
Full TCP inventory	22 and 80 open; 65,533 filtered	Small exposed service surface
Firewall scope	Anywhere to 10.10.10.0/24	Least-access improvement
Post-change state	22/80 open; 8080 filtered	Availability retained; blocked port unchanged
Evidence integrity	8 files returned OK	Artifacts unchanged after hashing

2. Objectives

- Discover live systems within the authorized SOC-Lab subnet.
- Inventory every TCP port on the Ubuntu target.
- Fingerprint exposed services and supporting web technology.
- Correlate remote observations with host-side listeners and service state.
- Reduce unnecessary firewall source exposure without disrupting authorized access.
- Preserve repeatable evidence and verify it with SHA-256.

3. Scope and Safety Controls

Control	Implementation
Authorization	Ryan Bryant's owned isolated VMs only
Assessment source	Parrot Security at 10.10.10.5
Primary target	LINUX-WEBSERVER01 at 10.10.10.4
Network boundary	10.10.10.0/24 SOC-Lab NAT network
Scan profile	SYN discovery, full TCP inventory, targeted scripts and versions
Excluded activity	No exploitation, password attacks, denial of service, or public scanning
Change safety	Add subnet rules before deleting broad rules; validate afterward

4. Assessment Workflow

01 CONNECTIVITY

Verified 10.10.10.4 responded with four packets and zero loss.

02 HOST DISCOVERY

Scanned the authorized /24 and documented four active addresses.

03 PORT INVENTORY

Scanned all 65,535 TCP ports with reasons and saved normal output.

04 SERVICE PROFILE

Ran default scripts and exhaustive version detection against TCP/22 and TCP/80.

05 PATH CONTEXT

Captured the Parrot interface, routing table, selected route, and ARP neighbor.

06 WEB PROFILE

Used WhatWeb to identify the HTTP platform and response controls.

07 CROSS-VALIDATION

Confirmed listening sockets and active SSH and Apache services on Ubuntu.

08 FIREWALL REVIEW

Compared exposed services with active UFW policy.

09 HARDENING

Restricted SSH and HTTP ingress to 10.10.10.0/24.

10 RETEST AND INTEGRITY

Rescanned ports 22, 80, and 8080, then verified eight evidence hashes.

5. Key Commands

```
sudo nmap -sn 10.10.10.0/24 -oN 01-subnet-host-discovery.txt
```

```
sudo nmap -sS -T3 -p 1-65535 --reason 10.10.10.4 -oN 02-full-tcp-port-inventory.txt
```

```
sudo nmap -sV --version-all -sC -T3 -p 22,80 --reason -oA 03-targeted-service-profile 10.10.10.4
```

```
whatweb -a 3 -v http://10.10.10.4 | tee 05-whatweb-technology-profile.txt
```

```
sudo ufw allow from 10.10.10.0/24 to any port 22 proto tcp
```

```
sudo nmap -sS -T3 -p 22,80,8080 --reason 10.10.10.4 -oN 06-post-hardening-exposure-validation.txt
```

SAFETY BOUNDARY All reconnaissance and configuration changes were confined to Ryan Bryant's isolated virtual lab and owned systems.

6. Evidence: Discovery and Full-Port Inventory

```
[x]-[user@parrot]~/SOC-Labs/Network-Reconnaissance
$ cat 01-subnet-host-discovery.txt
# Nmap 7.95 scan initiated Sun Aug 23 17:59:32 2026 as: nmap -sn -oN 01-subnet-host-discovery.txt 10.10.10.0/24
#
Nmap scan report for 10.10.10.1
Host is up (0.00080s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.10.10.2
Host is up (0.00068s latency).
MAC Address: 08:00:27:4D:F9:82 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 10.10.10.4
Host is up (0.00082s latency).
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 10.10.10.5
Host is up.
# Nmap done at Sun Aug 23 17:59:34 2026 -- 256 IP addresses (4 hosts up) scanned in 2.04 seconds
[user@parrot]~/SOC-Labs/Network-Reconnaissance
$
```

Figure 1. Host discovery identified four active lab addresses: .1, .2, .4, and .5.

```
[x]-[user@parrot]~/SOC-Labs/Network-Reconnaissance
$ sudo nmap -sS -T3 -p 1-65535 --reason 10.10.10.4 -oN 02-full-tcp-port-inventory.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-23 18:04 UTC
Nmap scan report for 10.10.10.4
Host is up, received arp-response (0.00079s latency).
Not shown: 65533 filtered tcp ports (no-response)
PORT      STATE SERVICE REASON
22/tcp    open  ssh      syn-ack ttl 64
80/tcp    open  http     syn-ack ttl 64
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 105.94 seconds
[user@parrot]~/SOC-Labs/Network-Reconnaissance
$ grep -E 'Nmap scan report|Host is up|^[0-9]+/tcp|Not shown|Nmap done' 02-full-tcp-port-inventory.txt
Nmap scan report for 10.10.10.4
Host is up, received arp-response (0.00079s latency).
Not shown: 65533 filtered tcp ports (no-response)
22/tcp    open  ssh      syn-ack ttl 64
80/tcp    open  http     syn-ack ttl 64
# Nmap done at Sun Aug 23 18:06:37 2026 -- 1 IP address (1 host up) scanned in 105.94 seconds
[user@parrot]~/SOC-Labs/Network-Reconnaissance
$
```

Figure 2. Full TCP SYN inventory found ports 22 and 80 open and 65,533 filtered.

7. Evidence: Service and Network Context

```

[user@parrot]--[~/SOC-Labs/Network-Reconnaissance]
$ sudo nmap -sV --version-all -sC -T3 -p 22,80 --reason -oA 03-targeted-service-profile 10.10.10.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-23 18:11 UTC
Nmap scan report for 10.10.10.4
Host is up, received arp-response (0.0012s latency).

PORT      STATE SERVICE REASON          VERSION
22/tcp    open  ssh      syn-ack ttl 64 OpenSSH 9.6p1 Ubuntu 3ubuntu13.18 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_   256 b2:cf:cf:8c:f1:4e:03:4a:01:98:bd:e3:d5:e4:a7:d1 (ECDSA)
|_   256 b1:e4:60:e2:55:45:68:98:04:de:5b:43:0b:db:d6:64 (ED25519)
80/tcp    open  http      syn-ack ttl 64 Apache httpd
|_ http-server-header: Apache
|_ http-title: Apache2 Ubuntu Default Page: It works
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.71 seconds
[user@parrot]--[~/SOC-Labs/Network-Reconnaissance]

```

Figure 3. Targeted service profiling identified OpenSSH 9.6p1 Ubuntu and Apache HTTP.

```

[user@parrot]--[~/SOC-Labs/Network-Reconnaissance]
$ cat 03-targeted-service-profile.nmap
# Nmap 7.95 scan initiated Sun Aug 23 18:11:22 2026 as: nmap -sV --version-all -sC -T3 -p 22,80 --reason -oA 03-targeted-service-profile 10.10.10.4
Nmap scan report for 10.10.10.4
Host is up, received arp-response (0.0012s latency).

PORT      STATE SERVICE REASON          VERSION
22/tcp    open  ssh      syn-ack ttl 64 OpenSSH 9.6p1 Ubuntu 3ubuntu13.18 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_   256 b2:cf:cf:8c:f1:4e:03:4a:01:98:bd:e3:d5:e4:a7:d1 (ECDSA)
|_   256 b1:e4:60:e2:55:45:68:98:04:de:5b:43:0b:db:d6:64 (ED25519)
80/tcp    open  http      syn-ack ttl 64 Apache httpd
|_ http-server-header: Apache
|_ http-title: Apache2 Ubuntu Default Page: It works
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done at Sun Aug 23 18:11:34 2026 -- 1 IP address (1 host up) scanned in 11.71 seconds
[user@parrot]--[~/SOC-Labs/Network-Reconnaissance]
$ ip -br address | tee 04-network-path-context.txt
lo        UNKNOWN    127.0.0.1/8 ::1/128
enp0s3    UP          10.10.10.5/24 fe80::58ac:aa7e:6033:4ef5/64
[user@parrot]--[~/SOC-Labs/Network-Reconnaissance]
$ ip route | tee -a 04-network-path-context.txt
default via 10.10.10.1 dev enp0s3 proto dhcp src 10.10.10.5 metric 100
10.10.10.0/24 dev enp0s3 proto kernel scope link src 10.10.10.5 metric 100
[user@parrot]--[~/SOC-Labs/Network-Reconnaissance]
$ ip route get 10.10.10.4 | tee -a 04-network-path-context.txt
10.10.10.4 dev enp0s3 src 10.10.10.5 uid 1000
cache
[user@parrot]--[~/SOC-Labs/Network-Reconnaissance]
$ ip neigh show 10.10.10.4 | tee -a 04-network-path-context.txt
10.10.10.4 dev enp0s3 lladdr 08:00:27:91:c9:2a STALE
[user@parrot]--[~/SOC-Labs/Network-Reconnaissance]
$

```

Figure 4. Parrot used a direct local-subnet path to the target; neighbor state STALE remained a valid cached mapping.

8. Evidence: Web and Host-Side Validation

```
[user@parrot:~/SOL-Labs/Network-Reconnaissance]
$whatweb -a 3 -v http://10.10.10.4 | tee 05-whatweb-technology-profile.txt
WhatWeb report for http://10.10.10.4
Status      : 200 OK
Title       : Apache2 Ubuntu Default Page: It works
IP          : 10.10.10.4
Country     : RESERVED, ZZ

Summary     : Apache, HTTPServer[Apache], UncommonHeaders[x-content-type-options,referrer-policy,content-security-policy,permissions-policy]

Detected Plugins:
[ Apache ]
    The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

    Google Dorks: (3)
    Website      : http://httpd.apache.org/

[ HTTPServer ]
    HTTP server header string. This plugin also attempts to identify the operating system from the server header.

    String       : Apache (from server string)

[ UncommonHeaders ]
    Uncommon HTTP server headers. The blacklist includes all the standard headers and many non standard but common ones. Interesting but fairly common headers should have their own plugins, eg. x-powered-by, server and x-aspnet-version. Info about headers can be found at www.http-stats.com

    String       : x-content-type-options,referrer-policy,content-security-policy,permissions-policy (from headers)

HTTP Headers:
    HTTP/1.1 200 OK
    Date: Sun, 23 Aug 2026 18:19:21 GMT
    Server: Apache
    X-Content-Type-Options: nosniff
    Referrer-Policy: strict-origin-when-cross-origin
    Content-Security-Policy: default-src 'self' 'unsafe-inline'; object-src 'none'; frame-ancestors 'none'; base-uri 'self'
    Permissions-Policy: geolocation=(), microphone=(), camera=()
    Last-Modified: Wed, 19 Aug 2026 19:34:52 GMT
    Accept-Ranges: bytes
    Vary: Accept-Encoding
    Content-Encoding: gzip
    Content-Length: 3121
    Connection: close
    Content-Type: text/html
```

Figure 5. WhatWeb confirmed Apache, HTTP 200, and the deployed security headers.

```
socstudent@linux-webserver01:~$ sudo ss -ltnp | grep -E 'State|:22|:80 '
[sudo] password for socstudent:
State Recv-Q Send-Q Local Address:Port Peer Address:PortProcess
LISTEN 0      4096      0.0.0.0:*        0.0.0.0:*        users:((("sshd",pid=942,fd=3),("systemd",pid=1,fd=93))
LISTEN 0      511      *:80            *:80            users:((("apache2",pid=2291,fd=4),("apache2",pid=2290,fd=4),("apache2",pid=950,fd=4))
LISTEN 0      4096      [::]:22        [::]:22        users:((("sshd",pid=942,fd=4),("systemd",pid=1,fd=94))
socstudent@linux-webserver01:~$ sudo systemctl is-active ssh apache2
active
active
socstudent@linux-webserver01:~$
```

Figure 6. Ubuntu confirmed SSH and Apache listeners and both services active.

9. Exposure Analysis

Observation	Risk / meaning	Disposition
TCP/22 SSH open	Expected administrative service; credential attacks are possible if broadly reachable	Retain for lab administration; restrict source subnet
TCP/80 HTTP open	Expected Apache service; default page reveals platform purpose	Retain for web labs; restrict source subnet and replace default content later
65,533 TCP ports filtered	Unsolicited traffic receives no response	Positive defensive posture
SSH/HTTP allowed from Anywhere	Broader than required inside the VM firewall	Remediated to 10.10.10.0/24
TCP/8080 denied	Existing block test port	Retained and validated filtered
Apache version hidden	Limits precise version intelligence	Retain current hardening

ANALYST JUDGMENT The broad Anywhere rules represented moderate configuration exposure inside the VM. Internet exposure was still constrained by the isolated NAT design, but subnet-scoping was the correct least-access improvement.

10. Firewall Hardening and Validation

```
socstudent@linux-webserver01:~$ sudo ufw allow from 10.10.10.0/24 to any port 22 proto tcp comment 'SOC Lab SSH Subnet'
Rule added
socstudent@linux-webserver01:~$ sudo ufw allow from 10.10.10.0/24 to any port 80 proto tcp comment 'SOC Lab Apache Subnet'
Rule added
socstudent@linux-webserver01:~$ sudo ufw delete allow 22/tcp
Rule deleted
Rule deleted (v6)
socstudent@linux-webserver01:~$ sudo ufw delete allow 80/tcp
Rule deleted
Rule deleted (v6)
socstudent@linux-webserver01:~$ sudo ufw status numbered
Status: active

    To      Action From
    --      -
[ 1] 8080/tcp DENY IN  Anywhere      # SOC Lab Block Test
[ 2] 22/tcp   ALLOW IN  10.10.10.0/24 # SOC Lab SSH Subnet
[ 3] 80/tcp   ALLOW IN  10.10.10.0/24 # SOC Lab Apache Subnet
[ 4] 8080/tcp (v6) DENY IN  Anywhere (v6) # SOC Lab Block Test

socstudent@linux-webserver01:~$
```

Figure 7. SSH and HTTP rules were restricted to 10.10.10.0/24 after broad rules were removed.

11. Post-Hardening Result

```

[user@parrot]--[~/SOC-Labs/Network-Reconnaissance]
$ sudo nmap -sS -T3 -p 22,80,8080 --reason 10.10.10.4 -oN 06-post-hardening-exposure-validation.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-23 18:35 UTC
Nmap scan report for 10.10.10.4
Host is up, received arp-response (0.0020s latency).

PORT      STATE      SERVICE      REASON
22/tcp    open       ssh          syn-ack ttl 64
80/tcp    open       http         syn-ack ttl 64
8080/tcp  filtered  http-proxy   no-response
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.41 seconds
[user@parrot]--[~/SOC-Labs/Network-Reconnaissance]
$ cat 06-post-hardening-exposure-validation.txt
# Nmap 7.95 scan initiated Sun Aug 23 18:35:58 2026 as: nmap -sS -T3 -p 22,80,8080 --reason -oN 06-post-hardening-exposure-validation.txt 10.10.10.4
Nmap scan report for 10.10.10.4
Host is up, received arp-response (0.0020s latency).

PORT      STATE      SERVICE      REASON
22/tcp    open       ssh          syn-ack ttl 64
80/tcp    open       http         syn-ack ttl 64
8080/tcp  filtered  http-proxy   no-response
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

# Nmap done at Sun Aug 23 18:35:59 2026 -- 1 IP address (1 host up) scanned in 1.41 seconds
[user@parrot]--[~/SOC-Labs/Network-Reconnaissance]
$

```

Figure 8. Authorized access to SSH and HTTP remained available; TCP/8080 remained filtered.

Control objective	Validation	Result
Preserve SSH access	TCP/22 from Parrot	Open
Preserve Apache access	TCP/80 from Parrot	Open
Retain denied test port	TCP/8080 from Parrot	Filtered
Reduce allowed source scope	UFW rule review	10.10.10.0/24 only

No exploitation or compromise was observed. The scan measured service exposure and validated defensive configuration only.

12. Evidence Integrity

```
[user@parrot]~/SOC-Labs/Network-Reconnaissance
$sha256sum \
> 01-subnet-host-discovery.txt \
> 02-full-tcp-port-inventory.txt \
> 03-targeted-service-profile.nmap \
> 03-targeted-service-profile.gnmap \
> 03-targeted-service-profile.xml \
> 04-network-path-context.txt \
> 05-whatweb-technology-profile.txt \
> 06-post-hardening-exposure-validation.txt \
> | tee SHA256SUMS.txt
e6c10a53f44ddc5bfc4d7ffab183c4a6122b41b2fafd99e289245c0be67319a1 01-subnet-host-discovery.txt
0ac9752b93ebdfb68c9e18ecaf29fb9f06c866b5b032b1dc7b54343648a649d4 02-full-tcp-port-inventory.txt
e3ca90e00dfef36c97d97212cdafb38e51262f8529193041fe315a6b595db398 03-targeted-service-profile.nmap
5eca9ffc07e583121de695a6b24d29a20809d6e02ff0b733b87b440f7208a1da 03-targeted-service-profile.gnmap
9f1d8455de6f5e98f7841ea6f411501646cecea0482394560bcabd930d4d8a58 03-targeted-service-profile.xml
eefd0a040fd8e70eec96352212e3764e27130df2a79ac1230b4f92afdeec0ccf 04-network-path-context.txt
6074a683d97b8bf510c01d9b68bfc3031918a53910327da64926ea4ac51523aa 05-whatweb-technology-profile.txt
f915039a30b3e2e1b92be0926574378a3726b6bb47352c6bec7af01aab2986a1 06-post-hardening-exposure-validation.txt
[user@parrot]~/SOC-Labs/Network-Reconnaissance
$sha256sum -c SHA256SUMS.txt
01-subnet-host-discovery.txt: OK
02-full-tcp-port-inventory.txt: OK
03-targeted-service-profile.nmap: OK
03-targeted-service-profile.gnmap: OK
03-targeted-service-profile.xml: OK
04-network-path-context.txt: OK
05-whatweb-technology-profile.txt: OK
06-post-hardening-exposure-validation.txt: OK
[user@parrot]~/SOC-Labs/Network-Reconnaissance
$ls -lh
total 36K
-rw-r--r-- 1 root root 654 Aug 23 17:59 01-subnet-host-discovery.txt
-rw-r--r-- 1 root root 547 Aug 23 18:06 02-full-tcp-port-inventory.txt
-rw-r--r-- 1 root root 415 Aug 23 18:11 03-targeted-service-profile.gnmap
-rw-r--r-- 1 root root 979 Aug 23 18:11 03-targeted-service-profile.nmap
-rw-r--r-- 1 root root 3.0K Aug 23 18:11 03-targeted-service-profile.xml
-rw-rw-r-- 1 user user 389 Aug 23 18:16 04-network-path-context.txt
-rw-rw-r-- 1 user user 2.0K Aug 23 18:20 05-whatweb-technology-profile.txt
-rw-r--r-- 1 root root 574 Aug 23 18:35 06-post-hardening-exposure-validation.txt
-rw-rw-r-- 1 user user 791 Aug 23 18:41 SHA256SUMS.txt
[user@parrot]~/SOC-Labs/Network-Reconnaissance
$
```

Figure 9. All eight evidence artifacts passed SHA-256 verification.

The evidence set includes host discovery, full-port inventory, three Nmap service-profile formats, network path context, WhatWeb output, and post-hardening validation. The manifest allows reviewers to detect later changes.

13. Skills Demonstrated

- Authorized subnet discovery and asset identification
- Complete TCP port inventory with Nmap SYN scanning
- Service and OS-family fingerprinting
- Network route and ARP-neighbor interpretation
- Web technology fingerprinting with WhatWeb
- Cross-source validation using host-side sockets and systemd
- UFW least-access hardening
- Post-change availability and exposure verification
- SHA-256 evidence integrity and technical reporting

14. Employer-Facing Project Summary

RESUME BULLET Conducted authorized subnet discovery and full TCP reconnaissance against an isolated Ubuntu server; correlated Nmap and WhatWeb results with host-side listeners, restricted SSH and HTTP access to the lab subnet with UFW, verified service availability, and preserved eight hashed evidence files.

15. Assistance and Source Transparency

This project was completed by Ryan Bryant in his personal home lab with assistance from OpenAI ChatGPT (Sarah). ChatGPT helped plan the authorized workflow, troubleshoot command syntax, interpret results, identify validation steps, locate official documentation, and organize this report. Ryan personally operated the VMs, entered the commands, changed the firewall policy, captured the evidence, and reviewed the conclusions. ChatGPT did not independently access or operate the lab computer.

16. References and Validation Sources

[Nmap Network Scanning - Host Discovery](#). Official host-discovery behavior and options.

[Nmap Port Scanning Techniques](#). TCP SYN scan interpretation.

[Nmap Service and Version Detection](#). Service fingerprinting behavior.

[Nmap Output](#). Normal, grepable, and XML evidence formats.

[WhatWeb official project](#). Web technology identification and aggression levels.

[Ubuntu UFW manual](#). Firewall rule syntax, status, and deletion.

FINAL DISPOSITION Authorized reconnaissance and service-exposure assessment completed. The exposed surface was documented, firewall access was narrowed, required services remained available, and eight evidence files were integrity-verified.